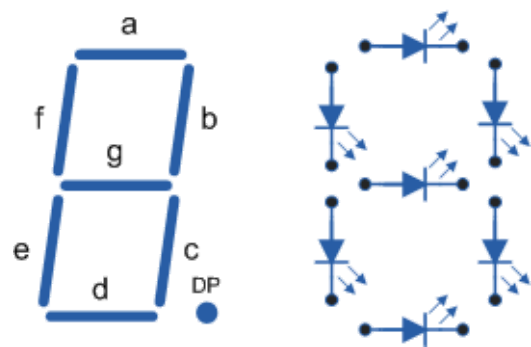


7-Segment Display Decoder Format



So in order to display the number “3” for example, segments a, b, c, d and g would need to be illuminated. If we wanted to display a different number or letter then a different set of segments would need to be illuminated. Then for a 7-segment display, we can produce a truth table giving the segments that need to be illuminated in order to produce the required character as shown below.

Truth Table for a 7-segment display

Individual Segments							Display
a	b	c	d	e	f	g	
x	x	x	x	x	x		0
	x	x					1
x	x		x	x		x	2
x	x	x	x			x	3
	x	x			x	x	4
x		x	x		x	x	5
x		x	x	x	x	x	6
x	x	x					7

Individual Segments							Display
a	b	c	d	e	f	g	
x	x	x	x	x	x	x	8
x	x	x	x		x	x	9
x	x	x		x	x	x	A
		x	x	x	x	x	b
x			x	x	x		C
	x	x	x	x		x	d
x			x	x	x	x	E
x				x	x	x	F

